

Business Requirement Document (BRD)

Project Title: Land Data Management Web Application

Version: 1.5

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1. Purpose

This document details the functional, structural, and workflow requirements of the Land Data Management Web Application, including land record uploads, data management, compensation workflows, document archival, audit tracking, and reporting.

2. Scope

The system supports:

- Upload and management of land records at Plot level.
- Hierarchical data navigation at Project → Village → Khata → Plot levels.
- Controlled editing & global data update propagation.
- Compensation processing via Payment Ready workflow.
- Document repository management per Khata.
- Deleted record retention.
- Unit conversion and tenant count auto-calculation.

3. User Roles

Role	Permissions
Admin	Full access including upload, edit, delete, restore deleted records, document upload, compensation workflows, reporting.
Data Entry User	Upload and edit records except Khata No. and Plot No., upload documents, cannot delete.
Viewer	Read-only access to permitted pages.

4. System Pages

1. Dashboard (TBD)
2. Private Land Module
 - Project Page

- Village Page
- Khata Page
- Plot Page
- Compensation Page
- Social Survey Page

3. Government Land Module

4. Forest Land Module

5. Import / Export Page
6. Reports Page
7. Deleted Records Page (Admin Only)

5. Data Hierarchy for Private Land Module

Project → Village → Khata → Plot

6. Data Upload Requirements for Private Land Module

Required Upload Fields

Field	Notes	Entry
Recorded Tenant(s)	Comma-separated allowed	User
Present Tenant(s)	Comma-separated allowed	User
Village Name		User
Village Code		User
Tahasil		User
RI Circle		User
Thana No.		User
Khata No.	(whitespace trimmed)	User
Plot No.	(whitespace trimmed)	User
Kissam		User
Land Category		User
Total Land Area (Acres)		User / System
Total Land Area (Hectares)		User / System
Land Area to be Acquired (Acres)		User / System
Land Area to be Acquired (Hectares)		User / System
LA Case File No.		User

Unit Conversion Logic

- If only **Acres** is provided → **Hectares = Acres × 2.471**
- If only **Hectares** is provided → **Acres = Hectares ÷ 2.471**
- If both provided → Use given values.
- Precision: **4 decimal places**

Data upload requirement for Government and Forest Land Module to be shared soon.

7. Unique ID Generation

Unique ID = Client Code / Village Code / Khata No.

8. Display Behavior by Page

Plot Page

Displays one row per plot with additional auto-calculated columns:

Additional Column Name	Behavior
Number of Recorded Tenants	Count = number of commas in Recorded Tenant(s) + 1
Number of Present Tenants	Count = number of commas in Present Tenant(s) + 1

Khata Page

- Groups plots by Unique ID.
- **Checkbox for selecting records**, plus **Select All**.
- **Document Upload Panel** appears per Unique ID.

Additional Column Name	Behavior
Completion of Survey	Manual Entry now. Automated Entry later on.
Completion of Yadast	Manual Entry now. Automated Entry later on.

9. Document Upload Repository

Location:

Khata Page, at the **Unique ID level**.

Upload Structure:

For each Unique ID, the system shall allow uploading **up to 3 files for each document type** listed below:

- | | |
|--|--|
| 1. Order Sheet | 16. Form 9A + Sample Photo (if any) |
| 2. Notice by GMDC | 17. Form 9B + Sample Photo (if any) |
| 3. Attendance Sheet | 18. Form 9C + Sample Photo (if any) |
| 4. Consent Form | 19. Land Acquisition Award |
| 5. Genealogy Sheet | 20. Indemnity Bond |
| 6. Legal Heir Certificate | 21. Physical Possession Certificate (Bond Paper) |
| 7. Yadast Register Copy | 22. Apportionment Affidavit (if applicable) |
| 8. Self-Attested RoR | 23. Affidavit for Legal Issues (if any) |
| 9. Certified Copy of RoR | 24. Aadhaar / Voter Card Copy |
| 10. Patta Original | 25. PAN Proof |
| 11. Encumbrance Certificate | 26. Bank Passbook / Cancelled Cheque Copy |
| 12. Rent Receipt | 27. Electronic Fund Transfer Form |
| 13. Trace Map | 28. Receipt of Compensation |
| 14. Application to Claim for Land Compensation | 29. Payment Voucher |
| 15. Calculation of Compensation | 30. Photo of Physical Possession |

GIS File Support (New Requirement)

- **Two (2) GIS files** may be uploaded and linked to the same Unique ID.

Metadata Tracking

Each uploaded file stores:

- Document Type
- Uploaded By (username)
- Upload Timestamp
- File Version Number (auto incremented per reupload)
- Option to View / Download

10. Editing, Deletion & Audit

(Delete moves to Deleted Records Page and does not destroy data.)

Global propagation: A change anywhere updates all affected pages and user is warned *before confirmation*.

Audit Fields on every record:

- Last Updated By
- Last Updated Timestamp

11. Compensation Workflow (Payment Ready)

Triggered by:

- Selecting Khata records on Khata Page and pressing **Payment Ready**.

Level-1 Header Row (One per Unique ID)

| Unique ID | Khata No. | Total Area to be Acquired (Ac / Ha) | Total Compensation Amount | Apportionment |

Level-2 Records (Auto-Generated)

One record is created **per Plot × Present Tenant**.

| Plot No. | Present Tenant | Payment Area | Compensation Payment | Apportionment (%) | Bank Account No. | Bank Name | IFSC Code | Payment Status | Transaction Number | File Upload (image/pdf) |

Validation Rules

- **Payment Area values must sum to Total Area to be Acquired.**
- **Compensation Payment values must sum to Total Compensation Amount.**

Color indicators:

Condition	Color
Values match expected total	Light Green
Values do not match	Light Orange

Apportionment Behavior

- Percentage entered → Payment auto-calculated.
- Manual Payment entry overrides percentage.

12. Import / Export

- Import template enforced.
- Export supports full dataset or filtered subsets.

13. Reports

Village, Khata, Plot, Tenant, Compensation, and Progress reports.

TBD

14. Non-Functional Requirements

Category	Requirement
Performance	Supports ≥50,000 rows per upload.
Security	RBAC, logs, attachment access control.
Consistency	Global update propagation + unit conversion integrity.
UX	Clean interface, consistent navigation, tooltip help for document uploads.

15. Future Enhancements

- GIS spatial mapping and land boundary overlays.
- Payment gateway integration for tracked compensation.